ANN ASSIGNMENT

Digit Recognition with LeNet-5 on MNIST

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GitHub repository: ann-assignment

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1 Dataset

The dataset used in this project is MNIST, a standard benchmark for handwritten digit recognition. The program loads it through `torchvision.datasets.MNIST` with separate training and test splits. In the code, the dataset contains 70,000 images in total:

- 60,000 training images
- 10,000 test images

Each image belongs to one of 10 classes, representing digits 0 through 9. Every sample is a 28×28 grayscale image, which the program converts into a 1×28×28 tensor for the CNN. If flattened, each image contains 784 pixel values.

Table 1: MNIST dataset summary

Property	Value
Total images	70,000
Training images	60,000
Test images	10,000
Number of classes	10
Image size	28×28 grayscale
Program input tensor	1×28×28
Flattened pixel values	784

2 LeNet-5 Model

LeNet-5 is one of the earliest and most influential convolutional neural network architectures for digit classification. In this project, it is adapted for MNIST by using one grayscale input channel and a padded first convolution so the spatial size remains 28×28 before pooling.

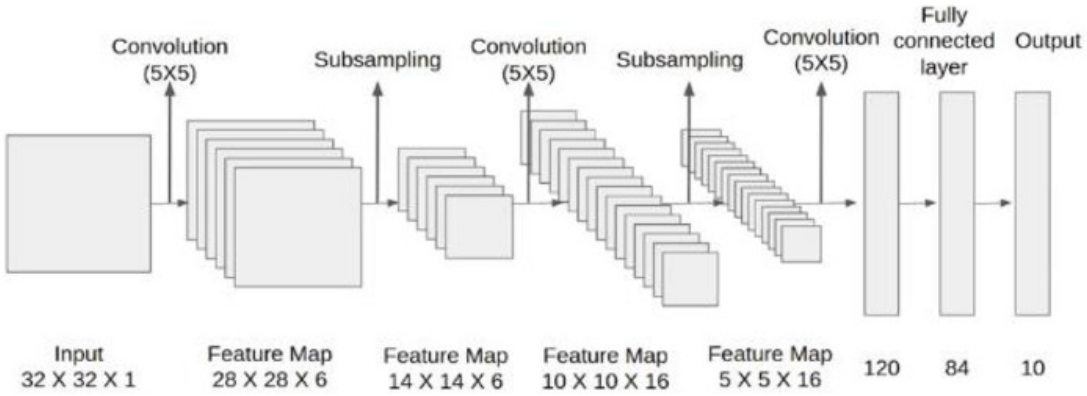

Figure 1: LeNet-5 CNN architecture reference used for the project report.

Table 2: Model architecture used in the project

Layer	Configuration / Output	Parameters
Input	1×28×28 grayscale image tensor	0
Conv1	6 filters, 5×5 kernel, padding 2, Tanh; output 6×28×28	156
Pool1	Average pooling, 2×2, stride 2; output 6×14×14	0
Conv2	16 filters, 5×5 kernel, Tanh; output 16×10×10	2,416
Pool2	Average pooling, 2×2, stride 2; output 16×5×5	0
Conv3	120 filters, 5×5 kernel, Tanh; output 120×1×1	48,120
Flatten	Reshape to a 120-dimensional vector	0
FC1	Linear 120→84, Tanh; output 84	10,164
Output	Linear 84→10 logits, one per digit class	850

The network has 61,706 trainable parameters. It is trained with Adam, batch size 64, learning rate 0.001, weight decay 1e-4, and a step learning-rate scheduler.

3 Observations

1. The model classified the clearer digits very confidently, with several predictions close to 100% confidence.
2. Digits 4 and 6 were more difficult because their handwriting styles were less distinct than the simpler digits.
3. The model confused 4 with 8 and 9 with 7, which shows that similar stroke patterns can lead to misclassification.
4. The preprocessing pipeline is important: grayscale conversion, inversion, cropping, padding, resizing, and normalization make custom images more consistent with MNIST.
5. Even with a compact size of 61,706 parameters, LeNet-5 achieved strong performance on handwritten digit recognition, which makes it a good lightweight CNN baseline.

Table 3: Inference results on custom images

Image	Actual class	Predicted class	Confidence
0.png	0	0	99.99%
1.png	1	1	99.99%
2.png	2	2	99.89%
3.png	3	3	100.00%
4.png	4	8	62.25%
5.png	5	5	99.98%
6.png	6	6	73.87%
7.png	7	7	98.86%
8.png	8	8	100.00%
9.png	9	7	99.95%

The custom image files are named by their true labels, so 0.png corresponds to class 0, 1.png to class 1, and so on.

The most useful curves for this project are the loss curve and the accuracy curve, because they show how quickly the model converged and how well it performed on held-out data. The plot below is generated from the real 30-epoch history saved in `lenet5_weights.json`. In this code, the MNIST test split is used for epoch-by-epoch evaluation, so these are test-set evaluation curves rather than a separate validation split.

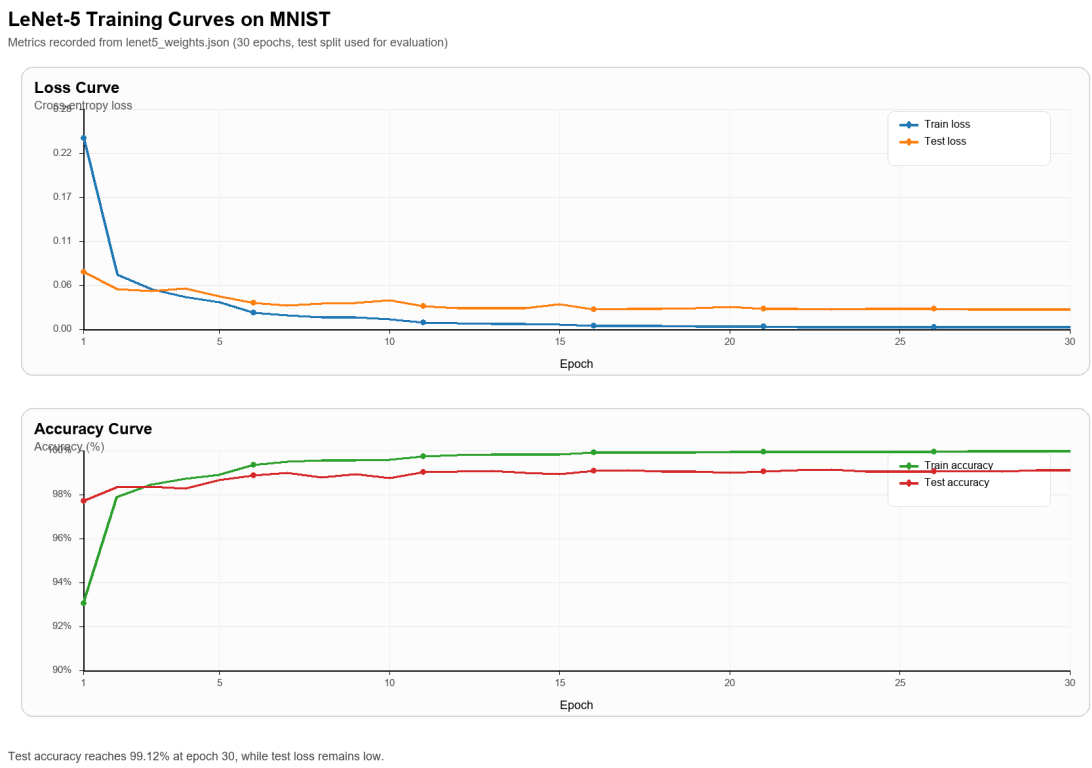

Figure 2: Training and test-set loss and accuracy over 30 epochs.

4 Conclusion

This project demonstrates that a LeNet-5 style convolutional neural network is effective for MNIST digit recognition. The dataset is simple enough for a compact model, yet it still reveals where ambiguity in handwriting can challenge the classifier. The results show that the model is highly reliable on most digits, while a few visually similar samples remain difficult cases.

The confusion matrix below is computed from the real 10,000-image MNIST test split using the trained model weights. It shows that most predictions fall on the diagonal, with only small off-diagonal errors between visually similar digits.

Confusion Matrix: MNIST Test Set

Rows = actual class, columns = predicted class. Counts come from the trained model on the real 10,000-image test split.

Actual class \ Predicted class	0	1	2	3	4	5	6	7	8	9
0	975	0	0	0	0	0	4	1	0	0
1	0	1130	1	0	0	1	2	0	1	0
2	0	1	1027	0	1	0	0	2	1	0
3	0	0	0	1003	0	3	0	1	3	0
4	0	0	2	0	974	0	2	0	0	4
5	1	0	0	6	0	883	2	0	0	0
6	4	2	0	0	2	1	948	0	1	0
7	1	3	6	0	0	0	0	1013	1	4
8	1	0	3	2	0	0	1	1	964	2
9	0	0	0	1	3	3	1	4	2	995

Overall test accuracy: 99.12% | Total test images: 10,000

Figure 3: Confusion matrix on the real MNIST test set, where rows are actual classes and columns are predicted classes.